

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 PRELIMS**

H2 BIOLOGY**9744/2****Paper 2 Mark Scheme**

Tuesday

12th September 2017

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

Conceptual error (CE)	Lack of Keywords (K)	Misreading the question (Q)

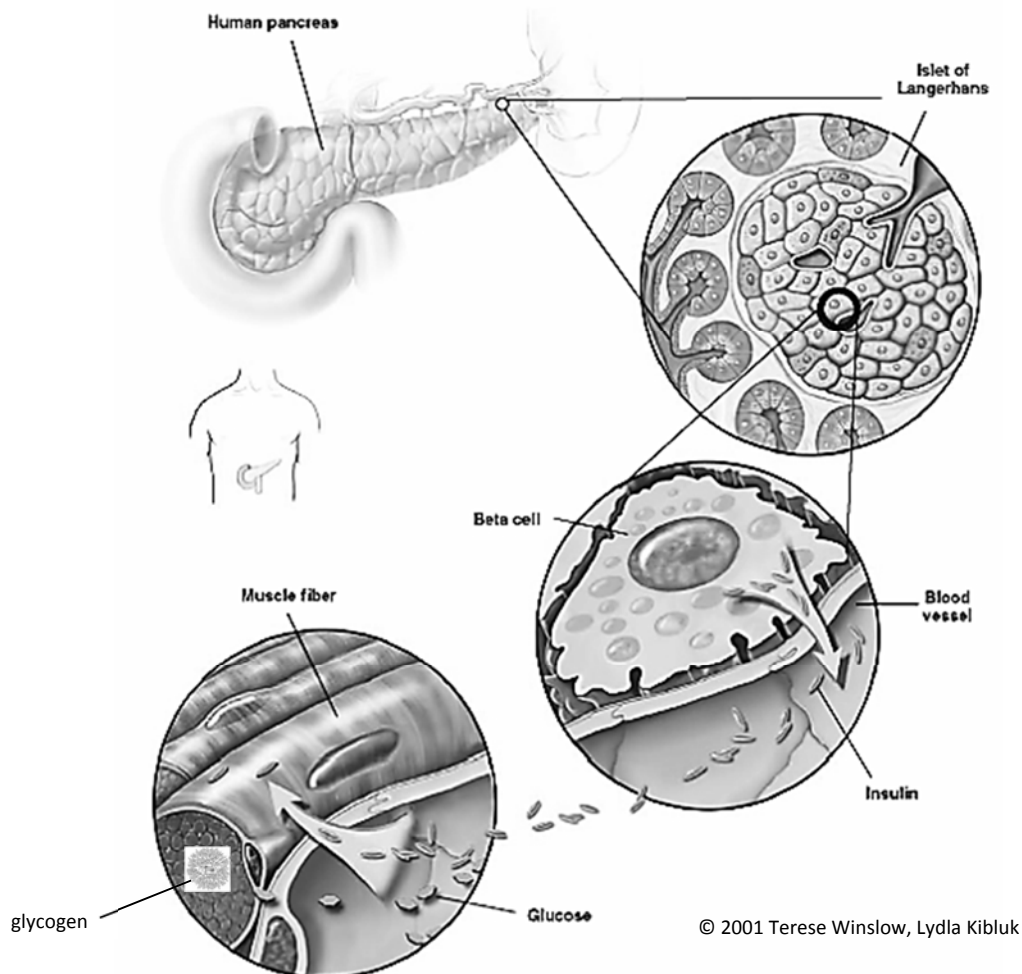
For Examiners' Use	
Section A	
1	/18
2	/14
3	/12
4	/12
5	/12
6	/12
7	/10
8	/10
Total	/100

This document consists of **xx** printed pages.

[Turn over]

QUESTION 1

Hormones, insulin and glucagon, are proteins that regulate the concentration of blood glucose level. Type 2 diabetes is characterized both by insulin resistance, a condition in which various tissues in the body no longer respond properly to insulin action, and by subsequent progressive decline in beta (β)-cell function to the point that the cells can no longer produce enough additional insulin to overcome the insulin resistance. Researchers are actively exploring use of stem cells as a potential source of deriving new β -cells to treat type 2 diabetes.



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Fig. 1

The pancreas is located in the abdomen, adjacent to the duodenum (the first portion of the small intestine). A cross section of the pancreas shows the islet of Langerhans which is the functional unit of the endocrine pancreas. Encircled is the beta cell that synthesizes and secretes insulin. Beta cells are located adjacent to blood vessels and can easily respond to changes in blood glucose concentration by adjusting insulin production.

- (a)** Cells that secrete proteins contain a lot of rough endoplasmic reticulum (rER) and a large Golgi body.

(i) Describe how the rER is involved in the production of insulin.

.....[1]

- 1 (RER has) **bound ribosomes** for protein synthesis

[REJECT: make amino acid]

[ACCEPT: amino acids joined together / polypeptide]

- 2 **Chemical** modification / post-translational modification of polypeptide

Note:

Point 2 is accepted in view that students have not learnt about the processing of insulin in detail. Chemical modification of insulin e.g cleavage of pro-insulin to insulin is done in the Golgi body

(ii) Describe how the Golgi body is involved in the secretion of insulin.

.....[2]

- 1 (Golgi body) further chemically modifies (insulin) ;

- 2 **packages** (insulin) into **secretory vesicles** which **move towards the cell surface membrane** (and fuse with it, to release insulin out of the cell) ;

(b) Using type II diabetes as an example, explain how environment affects phenotype.

.....[3]

- 1 people with functional pancreas/with no type I diabetes have **functional genes which code for insulin release**;
(insulin is secreted when blood glucose level increases);
- 2 **overeating** of sugary foods for a long period of time causes repeated stimulation of the pancreas;
which responds by **secreting high levels of insulin**;
- 3 **repeated exposure** of target cells to large amounts of insulin **desensitizes** the cells' responsiveness to insulin;
- 4 result in the target cells **failing to take in glucose**; (blood glucose stays high) resulting in type II diabetes;

(c) With reference to Fig. 1, explain how the binding of insulin to receptors on muscle cells leads to a lowering of blood glucose concentration.

.....[2]

- 1 Insulin binding results in the entry of glucose into the muscle cell by facilitated diffusion / via carrier (protein)/channel (protein)/glut4 protein/glucose transporter (down a concentration gradient)
- 2 Glucose used to form glycogen in the muscle cell (in Fig).

(d) Suggest how a change in the amino acid sequence of the receptor found in the plasma membrane of the muscle cell could make the cell resistant to insulin.

.....[2]

[Max 1]

- 1 Different amino acid sequence lead to different interactions between R groups of amino acids,
- 2 leading to **different tertiary structure / three-dimensional** structure (of receptor) ;

[Compulsory]

- 3 (so insulin) does not fit / bind / is not complementary ;
[REJECT: any reference to 'active site', 'enzyme-substrate complex' or insulin not fitting/binding to an enzyme]

(e) Describe how phospholipids are arranged in a plasma membrane.

.....[3]

- 1 (phospholipid molecules arranged as a) bilayer ; [ACCEPT : double layer]
- 2 **Polar phosphate** head / **charged** phosphate group (of phospholipid molecules) faces outwards and interacts with **aqueous** medium of the **external environment** and the **cytoplasm** ;
- 3 **Non-polar** hydrocarbon chains of fatty acids in phospholipid molecules form the **interior** of the plasma membrane / cell membrane / cell surface membrane ;

(f) Phospholipids are a type of lipid. Lipids, in general, are made up of glycerol and fatty acids monomers covalently bonded together. Name the covalent bond and describe the breakage of this bond.

.....[2]

- 1 ester bond ; [Reject: ester]
- 2 Addition of 1 water molecule across each ester bond (via hydrolysis reaction) ;
- 3 Products of hydrolysis are the **hydroxyl group (-OH) in the glycerol molecule** and the **carboxyl group (-COOH) of a fatty acid** ;

Experiments have indicated that pancreatic stem cells (PSCs) can serve as sources of insulin secreting cells.

(g) State the source of PSCs and explain the PSCs' normal functions.

.....[2]

- 1 Pancreas ;
- 2 Give rise to pancreatic cells, to growth, **repair and maintenance** of pancreatic tissues.

(h) Suggest an advantage of using the patient's own PSCs to regenerate tissue or organs.

.....[1]

- 1 No immune response (to own tissue)
/ tissue will not be rejected

[Reject: "cells will not be rejected" as context is on tissue regeneration]

[Q1: 18 marks]

QUESTION 2

Stephen Lyon Crohn, also known as "The man who can't catch AIDS", was a man notable for a genetic mutation, which caused his inability to contract AIDS.

Crohn had the "delta-32" mutation on the CCR5 co-receptor, a protein on the surface of cells involved in the immune system.

CCR5 "delta-32" mutation involves a 32-base-pair deletion in the CCR5 co-receptor gene locus.

(a) Describe the role of CCR5 co-receptor protein in the entry of Human HIV into host cells.

-[2]
1. Viral Gp120 **recognises and binds** to CCR5 co-receptor (in addition to CD4 receptor on T cell)
 2. thus, triggering an allosteric change in viral Gp41 which then pierces through the host plasma membrane
/ causing fusion between viral envelope and host plasma membrane

(b) Describe briefly how the information on DNA is used to synthesize the pre-mRNA transcript for CCR5 co-receptor.

-[3]
- 1 Ref. RNA polymerase II and General transcription factors / Transcription initiation complex bind to promoter sequence / TATA box; and **unzips** DNA helix ;
 - 2 adds complementary ribonucleotides to the 3' end of growing RNA chain ;
 - 3 ref. termination of transcription; cleaving occurs 10 to 35 nucleotides downstream of polyadenylation signal (of pre-mRNA) ;

Individuals can either gain full or partial resistance to HIV, depending on the number of mutated CCR5 alleles they possess at the CCR5 gene locus. Partial resistance to HIV occurs when the entry of the virus is severely hampered (but not completely prohibited).

Another Caucasian woman named Susan, is homozygous with the CCR5 "delta-32" mutation. Her husband has the same mutation on **one** CCR5 allele. The couple has a son and a daughter. Both son and daughter has 50% chance of having partial resistance and 50% chance of having full resistance.

(c) Explain if the inheritance of the CCR5 alleles is autosomal or X-linked.

-[2]
1. Autosomal;
 2. If autosomal, the mother / Susan will pass down the mutant allele to both son and daughter, the father will pass down either mutant or normal allele to son and daughter, children will either be fully resistant (homozygous) or partially resistant (heterozygous)
/ If X-linked, the mother / Susan will pass down the mutant allele to both son and daughter, the father will pass down mutant allele to only the daughter (as son inherits the Y chromosome), both son and daughter would be fully resistant

Explanation:

Key: allele A: normal allele; allele a: delta-32 mutant allele

If autosomal,

Mother has 2 delta 32 resistance alleles (aa)

Father has 1 susceptible allele and 1 resistance allele (Aa)

Son or daughter will either be aa (fully resistant) or Aa (50%)

If X linked,

Mother has delta 32 resistance allele ($X^a X^a$)

Father has 1 resistance allele and a Y chromosome ($X^a Y$)

Daughter: $X^a X^a$

Son: $X^a Y$

(d) Suggest why some individuals have partial resistance to HIV.

-[2]
1. Their genotype is heterozygous / they have one mutant allele and one normal allele;
 2. Expression of **both** the normal allele and the mutant allele/the normal allele and mutant allele are co-dominant to each other; leads to **both functional and non-functional** CCR5 co-receptor.

The son with **partial resistance** to HIV, marries a woman with **no resistance**.

The son has blood group O while his wife is heterozygous for blood group A.

(e) With the help of a genetic diagram, show all the possible outcomes for their offspring.

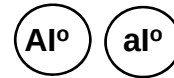
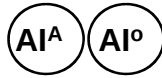
.....[5]

Key: "A": normal CCR5 allele

"a": mutant CCR5 allele

Parents phenotype: Wife No resistance Husband Partial resistance
parental genotype ;; AA I^A I^O x Aa I^O I^O

gametes;;



Punnett Sq showing offspring genotype

Gametes	AI ^A	AI ^O
AI ^O	AAI ^A I ^O No resistance, Blood gp A	AAI ^O I ^O No resistance Blood gp O
ai ^O	Aai ^O I ^A Partial resistance, Blood gp A	Aai ^O I ^O Partial resistance, Blood gp O

F1 phenotypic ratio:

1 no resistance, blood gp A: 1 no resistance, blood gp O: 1 partial resistance, blood gp A: 1 partial resistance, blood gp O

Mark scheme:

- 1 Parental genotypes
- 2 Parental gametes; circled
- 3 Punnett square showing all offspring genotypes
- 4 corresponds phenotypes to genotypes / legend for Punnett square
- 5 Offspring phenotypic ratio

[Q2: 14 marks]

Question 3

Epidermal growth factor (EGF) is released by cells, and is picked up either by the cell itself or by neighboring cells. It regulates the production of a number of proteins in target cells. Protein produced and its effect depends on the type of target cell.

Fig. 3 shows how EGF regulates 3 genes.

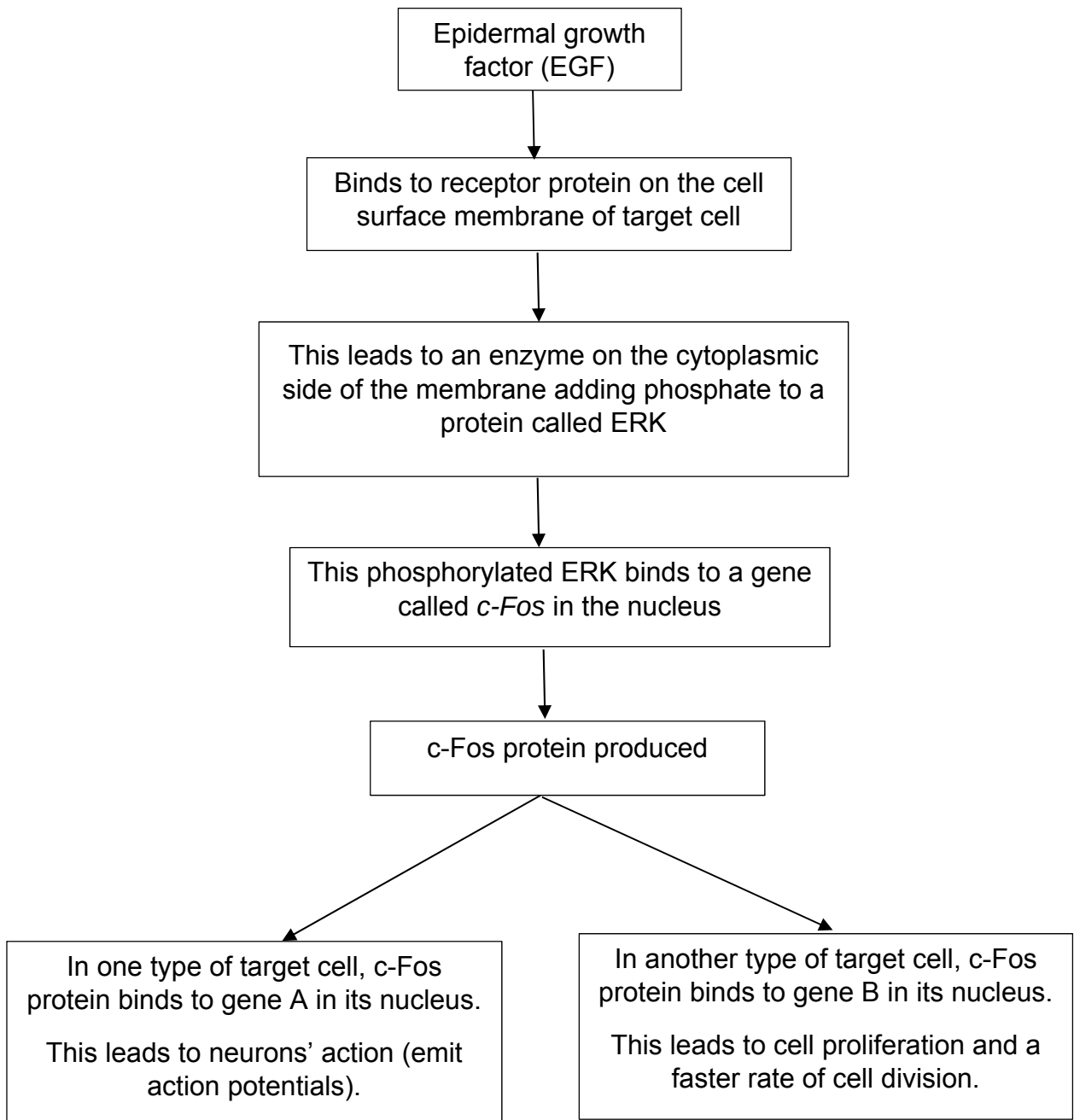


Fig. 3

(a) Name the **two** transcription factors in Fig. 3.

.....[1]

- 1 Phosphorylated ERK ; AND
c-Fos (protein)

(b) The *c-Fos* gene can be a proto-oncogene.

Use the information in Fig. 3 to explain how a mutation of the *c-Fos* gene can result in the formation of a tumour.

.....[3]

- 1 **gain of function** mutation of proto-oncogene to form oncogene
- 2 [Compulsory point] that code for **abnormal** c-Fos protein [Reject: overexpression] which is constitutively active / degradation-resistant;
- 3 lead to increased expression of gene B / **form more gene B product**, thus, **over-stimulation of the cell cycle** / cell keeps dividing [only award mark if point 2 is correct]

(c) Gene B has been associated with a significant number of human cancers. Scientists used polymerase chain reaction (PCR) to make multiple copies of gene B extracted from a patient's cancer tissue sample.

The reaction mixture includes the sample of DNA to be copied plus the following ingredients:

- DNA primers
- buffer solution
- heat-stable DNA polymerase (Taq polymerase)
- deoxyribonucleoside triphosphates (deoxyATP, deoxyTTP, deoxyCTP and deoxyGTP)

(i) Suggest why a buffer needs to be present in the reaction mixture.

.....[1]

- 1 to control the pH
/ to stop the polymerase denaturing
/ to optimise pH for polymerase activity

(ii) The deoxyribonucleoside triphosphates that are added to the reaction mixture are the monomers used for making the new DNA strands.

Suggest **one further** reason for adding the deoxyribonucleoside triphosphates to the reaction mixture.

.....[1]

- 1 *Ideas that it is a source of energy / AW ;*

(hydrolysis of the dATP to dAMP and PP release energy which is used in the catalysis of phosphodiester bonds in the polynucleotide chain)

- (iii) In the first stage of PCR, the mixture is heated to a temperature of around 90°C to denature the DNA. Suggest why high temperatures are needed to separate the two DNA strands.

.....[2]

- 1 *Idea of many hydrogen bonds between **complementary** strands together ;*
- 2 Hydrogen bonds break because of increased kinetic energy / vibrations ;

- (iv) At the end of several cycles of PCR, many copies of the DNA sample in the reaction mixture will have been made. The DNA samples are then separated out to produce a DNA banding pattern.

State the technique used to separate out the DNA samples **and** describe how this technique works.

.....[4]

- 1 Gel electrophoresis ;
- 2 (Load 10 µl of sample into the wells in agarose gel ;
Gel electrophoresis conducted at 100V till tracking dye move to $\frac{3}{4}$ length of gel)
DNA is **negatively-charged** (due to negatively-charged sugar-phosphate backbone) move towards the **positively-charged** electrode
- 3 through an agarose matrix which acts as a **molecular sieve** ;
- 4 DNA fragments separated by size ; where shorter DNA fragments move faster
[Reject: further] than longer ones;

[Q3: 12 marks]

QUESTION 4

Researchers have identified a gene that gives bacteria resistance to a type of antibiotics called polymyxins. Despite being discovered around 60 years ago, polymyxins maintained their effectiveness as antibiotics as they were seldom used due to concerns about their toxicity.

In recent years, rampant use of common antibiotics (e.g. penicillin) has led to the emergence of bacterial strains which are resistant to these antibiotics. This has become more and more of a global concern. Polymyxins are now a last line of defense against bacteria because of its previous lack of use.

(a) With reference to the reproductive cycle of bacteriophages, suggest how bacteriophage infections may lead to a spread of antibiotic resistance between bacterial populations.

-[3]
- 1 During generalised / specialised transduction, **host/bacteria DNA** can be incorporated into the phage capsid **randomly (for generalised transduction)/occasionally/by mistake** during viral assembly;
 - 2 The resulting transducing phages infect other bacteria and newly infected cell acquires the donor bacterial DNA
 - 3 Genetic recombination occurs and **expression** of antibiotic resistance genes result in phenotype of antibiotic resistance

Bacteria reproduce by the process of binary fission.

(b) Explain the significance of binary fission in bacteria

-[2]
- 1 Ref. asexual reproduction for unicellular organism
 - 2 Ensuring that offspring are genetically identical to the parent
/ Desirable alleles/traits are passed down
 - 3 Rapid increase in cell numbers (under favourable conditions)
- [Any 2]

The process of binary fission involves semi-conservative DNA replication.

(c) State two differences in the formation of the leading and lagging strands during DNA replication.

.....[2]

- 1 Presence of DNA ligase in lagging strand to ligate Okazaki fragments;
- 2 Presence of Okazaki fragments in lagging strand but none in leading strand;
- 3 Presence of more than 1 (RNA) primer/primase in lagging strand;
(**REJECT**: “no primer needed in leading strand”. This is incorrect!)
- 4 Strands are synthesized in opposite directions;
- 5 Leading strand is synthesized continuously vs lagging strand is synthesized discontinuously in the form of okazaki fragments

[Any 2]

Bacteria rely on sugar sources e.g. lactose for survival.

(d) Describe the consequence of mutating the *lacI* gene of the bacterial lac operon, on usage of lactose.

.....[5]

- 1 lac repressor has a change in 3D conformation at the **DNA-binding domain** / **allosteric site** so that allolactose / inducer binds tightly
- 2 lac repressor is inactive and is no longer able to bind to the operator (lacO) ;
- 3 RNA polymerase can constitutively access and transcribe the structural genes / lacZ , lacY and lacA;
- 4 β -galactosidase, lac permease and lactose transacetylase / (inducible) enzymes to utilize lactose are **constitutively** synthesised ;
- 5 Bacteria can utilize lactose even in the absence of lactose;

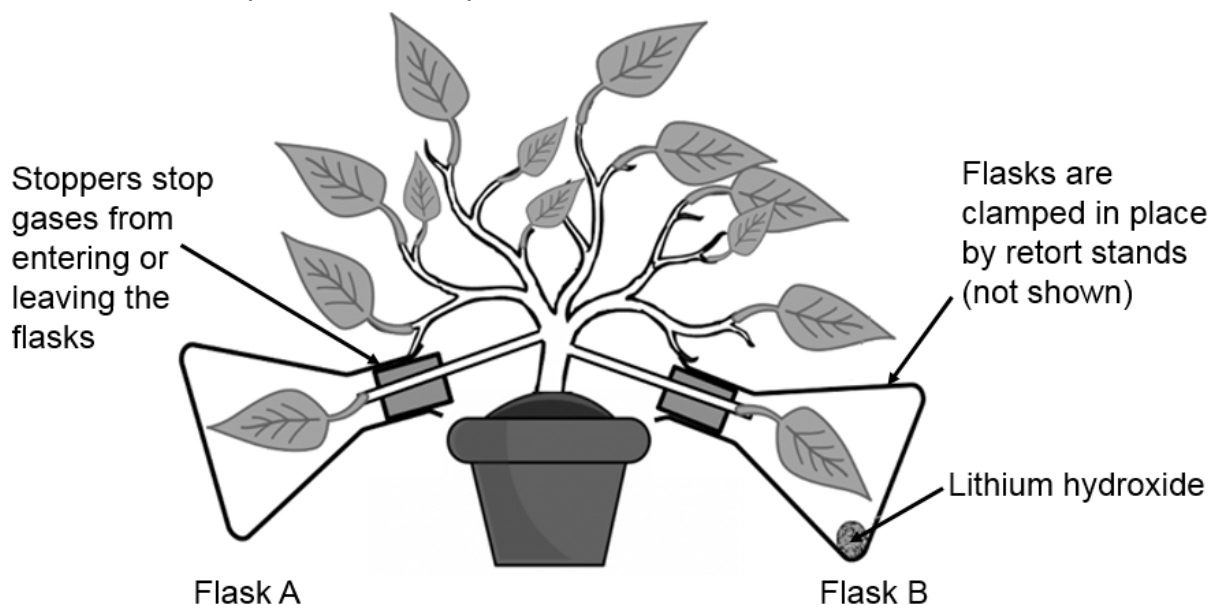
OR

- 1 lac repressor has a change in 3D conformation at the **DNA-binding domain** / **allosteric site** so that allolactose / inducer cannot bind
- 2 lac repressor (super repressor) binds tightly/continuously to the operator (lacO) ;
- 3 RNA polymerase cannot access and transcribe the structural genes / lacZ , lacY and lacA;
- 4 β -galactosidase, lac permease and lactose transacetylase / (inducible) enzymes to utilize lactose are **not** synthesised ;
- 5 Bacteria cannot utilize lactose even in the presence of lactose;

[Q4: 12 marks]

QUESTION 5

A student set up an experiment to investigate the effect of carbon dioxide on photosynthesis. First, he de-starched a small potted plant by putting it in the dark for two days. Then, he chose two leaves and inserted them into conical flasks, **A** and **B**, fitted with rubber stoppers. Lithium hydroxide was placed in Flask **B** to absorb all carbon dioxide present. The plant was then left under a table lamp for 15 minutes. Fig 5.1 shows the experimental setup.

**Fig 5.1**

He removed a sample of each leaf every 5 minutes (by punching out a leaf disc of approximately 0.2 cm in diameter, using a single-hole puncher) and return the leaves to their respective flasks immediately. Each leaf disc was then tested for the presence of ribulose biphosphate (RuBP) and starch. Table 5.2 shows the results he obtained.

Table 5.2

Flask	Concentration of RuBP / μmolm^{-2}				Concentration of starch / μmolm^{-2}			
	0 min	5 min	10 min	15 min	0 min	5 min	10 min	15 min
A	0.0	2.2	3.0	3.1	0.0	2.1	3.4	6.5
B	0.0	2.7	4.2	6.8	0.0	0.3	0.5	0.5

(a) State two other variables which must be kept constant to maximize the validity of the results obtained for this experiment.

.....[1]

- 1 Size of the leaves chosen (in flask A and flask B)
- 2 Distance from light source/light intensity
- 3 Temperature
- 4 Size of flask (affecting the volume of gas)

[Reject: leaves must be from the same plant as diagram already show only one pot of plants; pH of soil as fluctuations in pH will affect both leaves equally and hence, not affect the results]

(b) With reference to Table 5.2, describe the relationship between the presence of carbon dioxide and concentration of starch.

-[2]
- 1 In the presence of carbon dioxide (i.e. Flask A), starch concentration increased from 0.0 to 6.5 μmolm^{-2} from 0 - 15 min.
 - 2 while in the absence of carbon dioxide (i.e. Flask B), starch concentration increased much **less**, from 0.0 to 0.5 μmolm^{-2} from 0 - 15 min.

OR

- 1 The presence of carbon dioxide leads to a higher concentration of starch
- 2 E.g. at 15min, starch concentration in the presence of carbon dioxide (i.e. Flask A) is at 6.5 μmolm^{-2} while in the absence of carbon dioxide (i.e. Flask B), starch concentration is at 0.5 μmolm^{-2}

(c) Explain the absence of RuBP in both leaves at the start of the experiment.

-[2]
- 1 In the dark (during de-starching), the leaves carried out carbon fixation to convert **RuBP to PGA**.
 - 2 However, in the dark, no ATP and NADPH were produced,
 - 3 hence, chloroplasts are unable to convert **PGA to PGAL** and to regenerate **RuBP from PGAL**.

(d) The increase in RuBP concentration for the leaf in Flask A reached a plateau from 10 min to 15 min of exposure to light but continued to increase in the leaf in Flask B up to 15 min. Explain why.

-[3]
- 1 (In Flask A, concentration of RuBP reaches a plateau because) the RuBP that is being used up (in carbon fixation) equals the RuBP that is regenerated (from PGAL in the Calvin cycle).
 - 2 (In Flask B, exposure to light causes) the production of ATP and NADPH in the light-dependent reactions, which will convert existing PGA to PGAL (don't double penalize from part (c)), and **regenerate RuBP from PGAL**.
 - 3 However, in the absence of carbon dioxide, carbon fixation does not occur, and RuBP **accumulates** in the leaf.

The student watered the potted plant too excessively, causing the soil to become waterlogged. Fortunately, the roots of this plant could carry out anaerobic respiration under low oxygen conditions in the soil.

(e) Outline the process of anaerobic respiration in the roots under waterlogged conditions.

.....[4]

- 1 Glucose is converted to pyruvate via glycolysis in the cytosol.
- 2 Pyruvate is then decarboxylated to acetaldehyde/**ethanal** with the release of CO₂.
- 3 This reaction is catalysed by pyruvate decarboxylase.
- 4 NADH then reduces acetaldehyde/ethanal to ethanol,
- 5 catalysed by alcohol dehydrogenase.
- 6 NAD⁺ is **regenerated** for use in glycolysis to generate some **ATP** continuously.

[Q5: 12 marks]

QUESTION 6

The maggot fly, *Rhagoletis pomonella*, is native to the United States, and up until the discovery of the Americas by Europeans, fed solely on hawthorns. But when Europeans arrived on the Americas, they brought with them a new potential food source for the flies: apples.

At first, the flies are not attracted to apples. But over time, a minuscule change in the connections of two channels in the brain - one for detecting hawthorn odours and the other for apple odours - caused some flies to switch host plant (i.e. they jumped to apple trees). It was observed that the maggot flies strongly preferred to mate and lay their eggs on the type of tree they were born on.

When geneticists took a closer look in the late 20th century, they found that the two types - those that feed on apples and those that feed on hawthorns - have become genetically distinct groups.

(a) Identify the type of reproductive isolation shown in this case study.

.....[1]
 1 Behavioural isolation [Reject: Sympatric isolation; Geographical isolation]

(b) Discuss the processes that could have led to the genetic distinctions between the apple-eating flies and hawthorn-eating flies.

.....[5]
 1 (Behavioural isolation) **reduced the gene flow** between flies which jumped to apple trees and flies which remained on hawthorn trees.
 2 Processes of natural selection and genetic drift occurred.
 3 Different tree habitats exert different selection pressures
 4 Individuals which are better adapted to their respective environments **survive better, reproduce more and pass on their advantageous alleles to their offspring.**
 5 Different **mutations/genetic differences** accumulate in the different sub-populations of the flies. [Reject: speciation occurred]

(c) Suggest how researchers can conclusively determine whether the apple-eating flies are a different species from the hawthorn-eating flies.

.....[1]
 1 Observe for mating between apple-eating flies and hawthorn-eating flies, to produce viable and fertile offspring.

A nuclear small subunit (18S) rRNA gene has been sequenced for various insect species and used to reconstruct their evolutionary relationship, which is shown in a phylogenetic tree in Fig. 6.1.

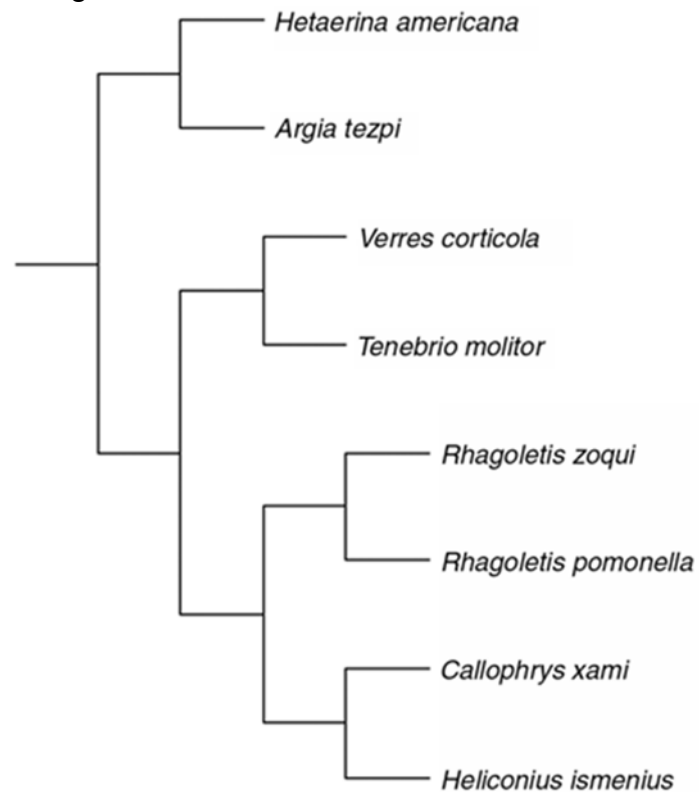


Fig. 6.1

(d) Comment on the evolutionary relationship between *Rhagoletis pomonella*, *Rhagoletis zoqui* and *Callophrys xami*.

-[1]
1. *Rhagoletis pomonella* and *Rhagoletis zoqui* are **more closely related** to each other **than either one is to *Callophrys xami***.
/ *Rhagoletis pomonella* and *Rhagoletis zoqui* share a **more recent common ancestor with each other than with *Callophrys xami***.

- (e) Discuss two advantages of using 18S rRNA gene over morphological comparisons for the reconstruction of the insect phylogeny.

.....[4]

Quantifiable and open to statistical analysis

- 1 Molecular data such as nucleotide and amino acid sequences are **quantifiable**, in abundance and **open to statistical analysis**. [Reject: the rRNA gene is in abundance]
- 2 Large quantities of data are required for statistical analysis; however there is **little morphological data available**.

Unambiguous and objective

- 3 Molecular data can be easily described in an **unambiguous/objective** [reject: **accurate**] manner.
- 4 Morphological data may be **subjective** and may differ depending on the way in which it was classified ~~In addition some characteristics may be analogous.~~

Not affected by convergent evolution

- 5 Molecular data provides a clear model of evolution by comparing the nucleotide and/or amino acid sequence as the **rate of molecular change in genes and proteins is regular** like a molecular clock.
- 6 Morphological evidence could be due to convergent evolution as similar morphology may not have been inherited from common ancestor /Rate of morphological change is not regular due to convergent or divergent evolution (hence cannot be used to reconstruct evolutionary relationships accurately)

Based strictly on heritable material

- 7 Molecular data is based strictly on **heritable** material.
- 8 Morphological data is based on anatomical characters which may be influenced by **environmental** factors as well as variation due to genotype of the organism.

Greater number of characters can be compared

- 9 DNA information provides **abundance of data** for analysis and it allows easy homology assessment.
- 10 Morphological **traits are few** and it is often difficult to assess homology for less complex structures.

[Any 2 pairs; no marks awarded if no attempt is made to compare between both molecular and morphological methods]

[Q6: 12 marks]

QUESTION 7

Influenza is a prevalent illness all over the world. Due to the high rate of change in the structure of the virus, annual vaccinations are recommended.

To overcome this **problem** of the constant need for developing new vaccines against influenza viruses, scientists are attempting to produce vaccines relating to antibodies which recognize part of the virus that does not change—the stalk of haemagglutinin. This research may lead to the development of “universal” influenza vaccines which can remain effective over long periods of time (Journal of Virology, 2017).

Fig. 7.1 shows the structure of haemagglutinin.

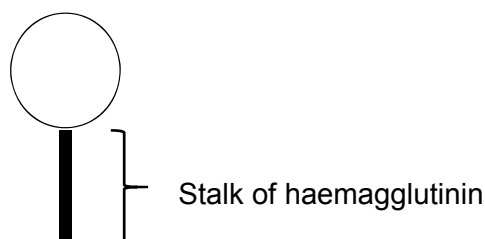


Fig. 7.1

(a) With reference to the context above, explain why there is a “constant need for developing new vaccines against influenza viruses”.

-[3]
1. Antigenic drift where there are spontaneous mutations in RNA genome coding for antigens haemagglutinin (and neuraminidase),
 2. due to **lack of proof-reading** in RNA-dependent RNA polymerase,
 3. change 3D conformation/tertiary structure of haemagglutinin (and neuraminidase),
 4. cannot be detected by **existing** memory B cells / antibodies present in the immune system, (thus same strain can infect the same person who is vaccinated previously)

OR

1. Antigenic shift where **two (or more)** different strains of the influenza virus infects the **same host cell**
2. There is **reassortment** of the viral RNA segments,
3. (giving rise to a new combinations of RNA segments in new viral particles, hence) **new combinations** of surface antigens haemagglutinin and neuraminidase arises; a new virus strain results,
4. cannot be detected by **existing** memory B cells / antibodies present in the immune system.

(b) With reference to the influenza virus and active immunity, define a vaccine and state how it provides protection against the virus.

-[2]
1. A vaccine contains antigen consisting of **dead/attenuated influenza virus/ parts of the influenza virus.**
 2. which stimulates artificial active immunity by **production of antibodies** by plasma cells/production of **memory cells after introduction/injection into body**; [Reject: vaccines produce antibodies]

Vaccinations can control diseases by resulting in *herd immunity*, in which large percentage of population which has become immune to the virus through vaccinations, provides a measure of protection for individuals who are not immune.

(c) Explain why.

-[1]
1. (When a large number of individuals are immune,) chains of infection are likely to be disrupted, which stops or slows the spread of disease.
/ The more people who are immune, the smaller the probability that those who are not immune will come into contact with an infectious individual.

Tuberculosis (TB) is an infectious disease that generally affects the lungs. Most infections are latent and do not have symptoms. Latent infections can progress to active form of the disease which, if left untreated, kills about half of those infected.

The most common classic symptom of active TB is chronic cough with blood-containing sputum.

(d) Name the organism which causes tuberculosis (TB).

-[1]
- 1 *Mycobacterium tuberculosis*

In the alveoli tissues, the bacteria that causes TB binds directly with mannose receptors on alveolar macrophages using a bacterial glycolipid, and is engulfed by alveolar macrophages.

(e) With reference to a cellular organelle in the macrophages, describe how macrophages attempt to process engulfed bacteria.

-[2]
1. Lysosomes contain **hydrolytic enzymes**, (e.g. proteases), which hydrolyzes bacteria
 2. after lysosomes **fused with phagosomes/endocytic vesicle containing bacteria**

(f) Describe the cause of the chronic cough symptom of TB in its **active** stage.

-[1]
1. Bacteria destroy alveoli/causes cavities in the lungs; this leads to **less surface area** for diffusion of gases.

OR

1. Damaged areas (may become infected with other bacteria) form pockets of pus; this **increases diffusion distance** between alveolar sac and alveolar capillaries

[Q7: 10 marks]

QUESTION 8

Proteaceae are a family of flowering plants predominantly distributed in the tropical and sub-tropical regions of Australia and South Africa. Threatened species may be vulnerable to climate change, because of their narrow temperature tolerance, small population sizes and restricted distributions.

A study modelled climate-induced changes on the range size (geographical distribution of a species) of 282 Proteaceae species. Figures 8.1a and 8.1b show the time course of range changes for wind-dispersed (**A**) and ant-dispersed (**B**) Proteaceae species (under 'full migration' dispersal assumptions, which assumes no limitation to migration). Error bars represent standard error (for example, smaller standard errors indicates that the observations are closer to the actual value).

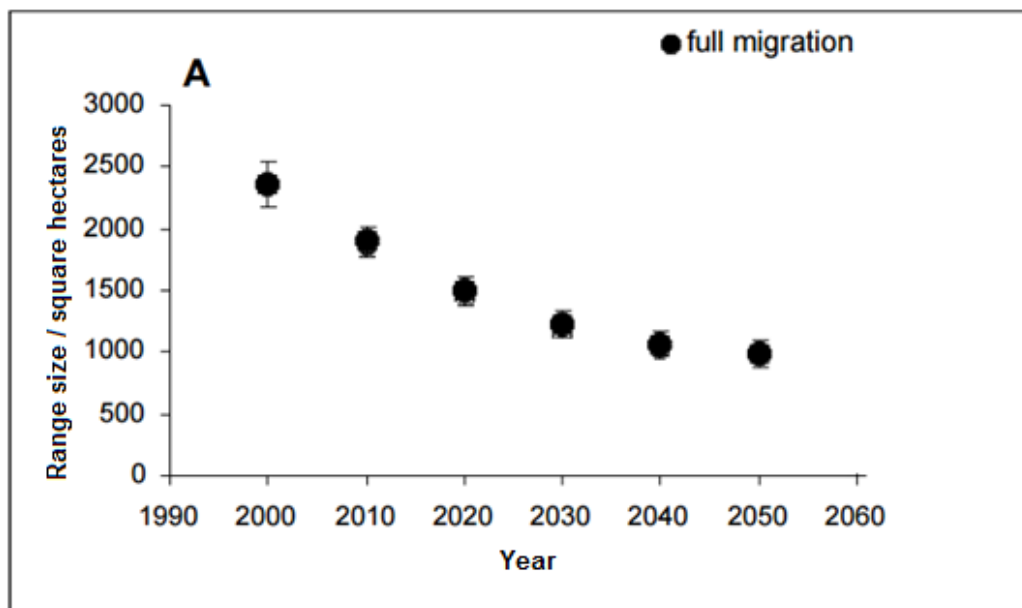


Fig. 8.1a

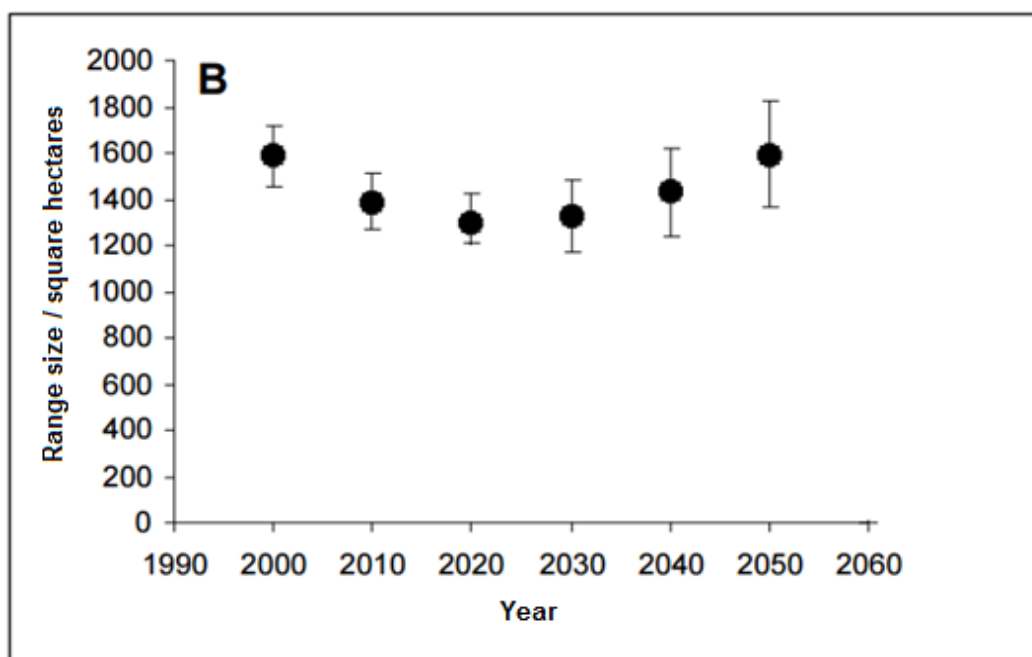


Fig. 8.1b

- (a) Describe the differences in climate-induced changes in range size between the wind-dispersed and ant-dispersed Proteaceae species from Year 2000 to 2050.

.....[2]

- 1 The range size for wind-dispersed Proteaceae species **decreases** from 2350 square hectares (accept 2300 – 2400) in year 2000 to 1000 square hectares in year 2050 while
- 2 the range size for ant-dispersed species **decreased** from 1600 to 1300 square hectares from year 2000 to 2020 but **increased back** to 1600 square hectares in year 2050.

- (b) The Intergovernmental Panel for Climate Change (IPCC) predicted an increase of 2°C in global temperatures by 2050. Suggest reasons for the larger range size of ant-dispersed Proteaceae species compared to that of wind-dispersed species in 2050.

.....[4]

[Ant-dispersed species]

- 1 Increase in temperature increases rate of metabolism / development of insects such as ants

Explanation for students' understanding:

Higher rate of metabolism causes ants to be more physically active and travel further distances in a day;

Higher rate of development of insects leads to eggs hatching and developing into adults faster, thus, increasing population size faster. The larger the ant colony size, the larger the foraging range

- 2 This increases the **distance/area** over which the **ants can disperse the seeds** of the ant-dispersed Proteaceae species.

OR

- 1 Increase in temperature may cause some ant colonies to migrate to higher latitudes/cooler regions
- 2 This increases the **area** over which the **ants can disperse the seeds** of the ant-dispersed Proteaceae species.

[Reject: This increases the range size of ant-dispersed species]

AND

[Wind-dispersed species]

- 3 The increase in temperature may result in **changes in wind patterns** (e.g. less wind / wind is less strong),
- 4 This restricts the **area** over which the wind can disperse the seeds of wind-dispersed species.

OR

- 3 Plants also adapt to increased temperatures by reducing the number of stomata per leaf (to reduce amount of water lost via evapo-transpiration).
- 4 The rate of photosynthesis decreases, and the populations of wind-dispersed species may reduce in number and hence range size.

It was predicted that not only will the range size for ant-dispersed Proteaceae species remain high in 2050, it is likely that the plants would colonise new regions, particularly in the higher latitudes and altitudes.

(c) Explain why the Proteaceae species can potentially threaten the native species of the regions they expand into.

.....[2]

- 1 Proteaceae species may become invasive species which **compete** more successfully with the native species for resources such as nutrients, sunlight and space (name at least one).
- 2 The **absence of natural predators** / herbivores which can feed on the new Proteaceae species will allow the plants to grow unchecked / more vigorously than native species (which have natural predators that control their population size).
- 3 [only award mark if point 1 or 2 is listed] This can cause the extinction of native species in these regions / the **survival** of native species is affected

Many indigenous cultures in tropical regions have used Proteaceae for medicinal preparations. For example, bioactive ingredients can be obtained from infusions of the roots, bark, leaves, or flowers of many Proteaceae species, which can be used as topical applications for skin conditions or internally as tonics, aphrodisiacs, and medicines to treat headaches, cough, diarrhea, indigestion, stomach ulcers, and kidney disease.

(d) Discuss how climate change can affect the biodiversity of the Proteaceae species and its consequence to the production of biomedicines.

.....[2]

- 1 Tropical species are often sensitive to hot weather as many species have evolved to become **thermal specialists** / have a narrow temperature tolerance.
- 2 Hence they are very **vulnerable** / cannot adapt to **climate warming** [Reject: climate change as it is lifted from question stem]
- 3 With the loss of biodiversity due to climate changes, production of biomedicines decreases / potential uses of Proteaceae species for biomedical usage may be **lost forever**.

[Q8: 10 marks]